Advanced Prompting and Intro to Langchain

Wednesday, February 18, 2026 6:00 PM

Recap: Customizing llm generation

1) ✓ Controlling the parameters
Temp, top-p and top-k

2) prompt engg → Adv

3) Prompt Chaining → Detailed Instru

4) Fine-tuning ≈
↳ amt of data
↓
Incorporate kn

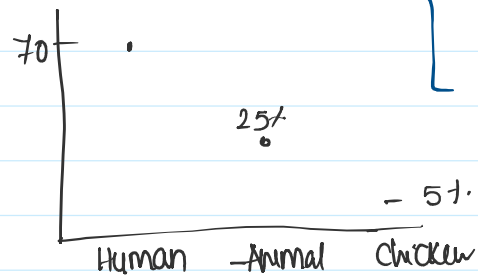
RAG

Temperature:
→ Scales the prob
↓
prob of next word in the generation

I am a _____

predict Next Token

By def
- Temp = 1
- top-p
- top-k



→ LLM → go with strategy = 'Greedy' → Always pick Next word as high prob

Temp - 0
↓
Scales the prob
↳ only highest prob word remains

I am a human
↳ Boring Content

ex: Human - 100 %
Animal - 0 %
Chicken - 0 %

Generate text acc to Training Content

Zero Creativity Guess

Randomly pick the next word from list X
↓
Scale the prob
↳ I am a chicken
↓
Wrong content of

Scale the prob

↓

Temp.

↓

0 ⟷ 2

1 → Same prob

2 → Scales the prob of low prob words in the list

↓

The chance of picking other words increases

↓

Creative Content Generation

↓

whole content of generation changes.

Temperature — Creativity knob.

top-K

LLM ↓

↳ predict next word as a word from top-K high prob words

I am a _____

Human 70%. ①

Animal 25%. ②

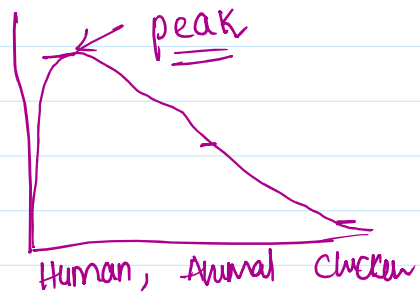
Chicken 5%.

top-K = 2 ⇒ as a next word pred

LLM — equal chance of picking Human/Animal

↳ Quite suitable

Next word prob distrib^n

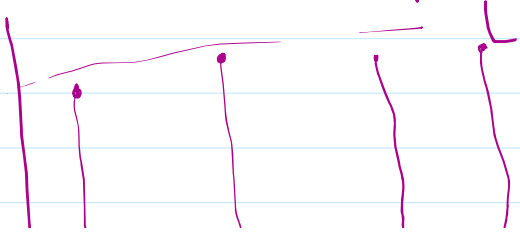


top-P (Freq used along with Temp)

↳ Suitable ⇒ Next word prob dist ≈ Uniform distrib^n

I am

0.2



I am

⇒ p- Cum prob.

↳ Select least number of words.

top-p = 0.9.

↳ $P_{w_1} + P_{w_2} + P_{w_3} \gtrsim 90\%$

↓

Sign #1 words.

↓

All words equal chance by LLM as next word pred.

Prompt Engg

↓

Articles

↓

Different Manner

1) Word precision

2) Balance length

3) Roles. System – LLM- Teacher
User – User- Student

↳ personas.

4) Scope

5) positive & Negative

↓ Include ↓ Exclude

Adv prompt Engg

↳ Simple prompts → will fail

↳ LLM- understand User Intention with sample prompts

LLM's are Lazy thinkers

↳ Make LLM to think → Think prompt